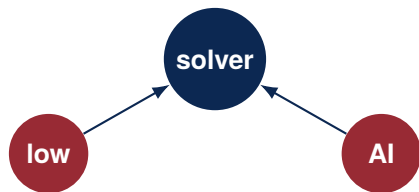


20-MINUTE PRESENTATION · REPOSITORY 05

AI in the Knowledge Economy

Capability, autonomy, and who captures the surplus



John Barraza · Universidad del Pacífico

github.com/johnbarraza/ai-05-ide

Roadmap and contribution

Paper

Organization: workers, solvers, and scalable AI.

Outcomes: matching, wages, output.

Comparison: autonomous versus co-pilot-only AI.

Claim tested

“Distribution is driven by autonomy, not capability.”

My audit

I recover the exact conditions in Propositions 5–6, then use a three-type zero-profit calculation to show why capability and autonomy are separate margins.

The economic question

Question

How do AI capability and autonomy reorganize knowledge work, total output, and the distribution of labor income?

Capability

How difficult a problem AI can solve: Z_{AI} .

Autonomy

Whether AI can pursue opportunities, not merely advise.

Organization

Who works, who solves, and each solver's span of control.

The paper's contribution is the interaction of all three—not a one-dimensional treatment taxonomy.

Primitives

- Human knowledge $z \in [0, 1]$, perfectly observable.
- Distribution G has continuous, strictly positive density g .
- Each person owns one unit of time and maximizes income.
- Problem difficulty $x \sim U[0, 1]$; success iff $z \geq x$.
- Asking a solver consumes $h \in (0, 1)$ units of solver time.
- Competitive, zero-fixed-cost firms; at most two layers.
- AI knowledge $z_{AI} \in [0, 1]$; one unit of compute per agent.
- Compute is abundant relative to human time.

Maintained later

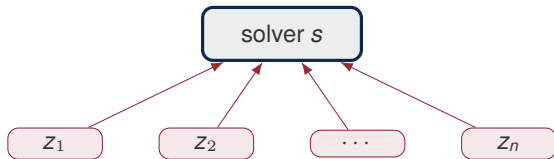
The distributional section restricts attention to $h < h_0$: no independent human producers pre-AI.

The hierarchy technology

A worker with knowledge z solves fraction z of problems; fraction $1 - z$ reaches the solver. One request costs h , so a solver can support

$$n(z) = \frac{1}{h(1 - z)}$$

identical workers.



Lower h or higher worker knowledge increases span of control.

The firm's problem

For a human hierarchy, a firm hiring workers z chooses a solver $s \geq z$:

$$\max_{s \geq z} \Pi_2^{nA}(s, z) = n(z)[s - w(z)] - w(s).$$

With autonomous AI it also compares

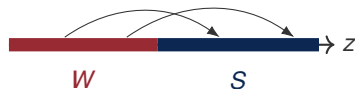
$$\begin{aligned} \Pi_1^A &= z_{AI} - r, & \Pi_2^{tA}(z) &= n(z)[z_{AI} - w(z)] - r, \\ \Pi_2^{bA}(s) &= n(z_{AI})[s - r] - w(s). \end{aligned}$$

Equilibrium discipline

Firms choose the profit-maximizing structure, free entry sets active profits to zero, labor markets clear, and compute earns rent r .

Pre-AI benchmark

- Occupational stratification: $W \preceq I \preceq S$.
- Positive assortative matching: $m : W \rightarrow S$ increases.
- W and S are nonempty; I exists iff $h > h_0$.
- Wage $w(z)$ is continuous, increasing, and convex.



For $h < h_0$, everyone belongs to a two-layer organization.

What counts as autonomous AI?

One unit of compute creates one agent of knowledge z_{AI} , a perfect substitute for a same-knowledge human.

Autonomous

AI may act as:

- independent producer;
- worker/co-worker;
- solver/co-pilot.

Non-autonomous

AI may act only as solver/co-pilot. It cannot pursue production opportunities itself.

Autonomy is a constraint on roles; capability is the location z_{AI} on the knowledge axis.

Why abundant compute fixes the autonomous rent

Some autonomous AI must produce independently when compute is abundant relative to human time. Zero profit then implies

$$\Pi_1^A = z_{AI} - r^* = 0 \implies \boxed{r^* = z_{AI}}.$$

Opportunity cost

An AI agent used as solver forgoes independent output z_{AI} .

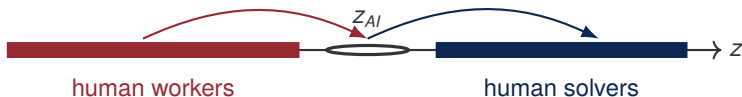
Distribution

That outside option affects how much surplus the assisted human workers can retain.

Autonomous AI reorganizes both sides

Occupational stratification places workers weakly below AI and solvers weakly above it:

$$W^* \preceq \{z_{AI}\} \preceq S^*.$$



AI can displace a match and also change the division of its surplus.

Two wage channels

Match effect

AI changes the knowledge of the partner assigned to a human. At both extremes, the relevant rematching can depress wages.

Share effect

AI changes the partner's price and therefore the human's share of team surplus. This effect can raise wages.

$$\Delta w(z) = \text{match effect} + \text{share effect}$$

At the bottom, the sign depends on AI capability. At the top, low communication cost makes the positive share effect dominate.

Proposition 5: define the objects first

Compare the autonomous wage w^* with the pre-AI wage w :

$$B \equiv \{z \in [0, z_{AI}] : w^*(z) > w(z)\},$$

$$T \equiv \{z \in [z_{AI}, 1] : w^*(z) > w(z)\}.$$

The proof shows these sets have cutoff form:

$$B = [0, z_b), \quad T = (z_t, 1]$$

for some $z_b \geq 0$ and $z_t \leq 1$.

Interpretation

If autonomous AI produces wage winners, they lie at the extremes of the knowledge distribution—not necessarily at both extremes.

Proposition 5: exact statement and conditions

Bottom

There are winners at the bottom iff AI is good enough:

$$B \neq \emptyset \iff z_{AI} > \bar{z}_{AI}, \quad \bar{z}_{AI} \in \text{int}(W).$$

Top

There are winners at the top:

$$T \neq \emptyset \quad \forall z_{AI} \in [0, 1).$$

Maintained: continuous positive density, two layers, abundant compute, $h < h_0$, and $z_{AI} < 1$. The paper notes the top claim may fail for $h \geq h_0$.

Source: Proposition 5 and footnote 19, accepted manuscript, pp. 25–26.

Why bottom winners require capability

For a basic AI, $z_{AI} \in \text{int}(W)$:

Negative match effect

The least knowledgeable lose access to their former, more knowledgeable human solver.

Positive share effect

AI can be cheaper than the displaced human solver, increasing the workers' share.

The positive effect dominates only above $\bar{z}_{AI}$. When $z_{AI} \in \text{int}(S)$, both match and share effects at the bottom are positive.

Key

The existence of bottom winners is explicitly a capability result.

Why the top result is conditional

The most knowledgeable also face a negative match effect and positive share effect. Under $h < h_0$, team size is large, so a small fall in worker cost is amplified across many workers:

$$n(z) = \frac{1}{h(1-z)}.$$

Do not overstate Proposition 5

- When $h \geq h_0$, the paper says top humans can sometimes lose.
- The theorem uses $z_{AI} < 1$; at $z_{AI} = 1$, the most knowledgeable necessarily lose.

“There are always winners at the top” means always inside the maintained parameter region.

Non-autonomous AI changes one constraint

AI can solve but cannot pursue a production opportunity. Therefore:

$$S_a = \emptyset, \quad \mu_w = 0,$$

and some compute remains idle. With abundant compute,

$$r^* = 0.$$

Same capability

Knowledge remains z_{AI} .

This zero rent is the source of a stronger share effect for low-skilled users.

Different outside option

A co-pilot cannot earn z_{AI} independently.

Proposition 6: equilibrium and adoption

There is a unique efficient equilibrium that maximizes labor income, with $r^* = 0$.

$$z_{AI} \leq w(0)$$

AI is not used: $W_a^* = \emptyset$. Human occupations and wages coincide with pre-AI outcomes.

$$z_{AI} > w(0)$$

Only the least knowledgeable use AI as solver:

$$W_a^* \preceq (W_p^* \cup I^* \cup S_p^*),$$

with all sets nonempty except possibly I^* .

The adoption cutoff $z_{AI} > w(0)$ is another capability condition.

Proposition 6.1: output ranking

Holding compute availability identical and abundant, and maintaining at most two layers,

$$Y^* > Y^*.$$



This is an aggregate output comparison, not a Pareto ranking of human wages.

Proposition 6.2: non-autonomy creates losers

In all cases,

$$\exists z \in (0, 1] \quad w^*(z) \leq w(z),$$

with strict inequality if $z_{AI} > w(0)$.

Logical detail

When $z_{AI} \leq w(0)$, AI is unused and everyone is weakly unchanged. When it is active, some humans strictly lose.

Efficiency and maximum labor income do not imply every type gains; organizational rematching redistributes labor income.

Proposition 6.3: the bottom prefers non-autonomy

There exists $\varepsilon > 0$ such that for every $z \in [0, \varepsilon)$,

$$w^*(z) \geq \max\{w(z), w^*(z)\},$$

strictly if $z_{AI} > w(0)$.

Mechanism

Co-pilot-only AI does not compete with low types for production and has zero compute rent. Low-skilled workers capture more of the automated team's surplus.

But the strict gain still switches on only when capability crosses $w(0)$.

Proposition 6.4: the top prefers autonomy

There exists $\varepsilon > 0$ such that for every $z \in (1 - \varepsilon, 1]$,

$$w^*(z) \leq w^*(z),$$

strictly for $z \neq 1$.

Mechanism

The most knowledgeable benefit when autonomous AI can work below them and expand their span of control. Co-pilot-only AI removes that channel.

Thus output and top wages favor autonomy, while bottom wages can favor non-autonomy.

The correct two-dimensional taxonomy

	Low capability	High capability
Autonomous	Top winners; bottom need not win	Bottom winners once $z_{AI} > \bar{z}_{AI}$; top winners under $h < h_0$
Non-autonomous	Unused if $z_{AI} \leq w(0)$	Active solver for the bottom; bottom strictly prefers it

Verdict on the slogan

Autonomy determines feasible roles and surplus channels. Capability determines which cells are actually reached.

Two-type hand derivation

Let $L < H$, $a = z_{AI}$, $n_L = [h(1 - z_L)]^{-1}$, and pre-AI wage w_L^0 .

Autonomous: $r^* = a$

$$n_L(a - w_L^A) - a = 0$$

$$w_L^A = a[1 - h(1 - z_L)].$$

Thus L wins iff

$$a > \bar{a}_L = \frac{w_L^0}{1 - h(1 - z_L)}.$$

Non-autonomous: $r^* = 0$

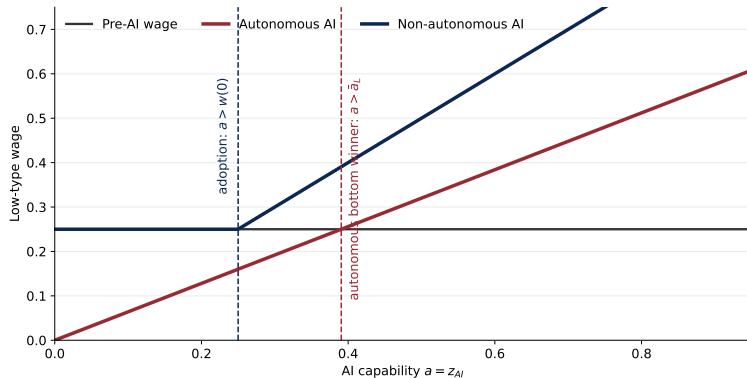
Conditional on $a > w(0)$,

$$n_L(a - w_L^N) = 0 \Rightarrow w_L^N = a,$$

so

$$w_L^N - w_L^A = ah(1 - z_L) > 0.$$

Reproducible numerical check



For $(z_L, h, w_L^0) = (.10, .40, .25)$:

$$w_L^A = .64a, \quad \bar{a}_L = .390625.$$

At $a = .30$, non-autonomous AI is active but autonomous AI does not make L a winner. At $a = .60$, both do; L still prefers non-autonomy.

Source: Generated by discrete_model.py; equations verified symbolically with SymPy.

Lean check: what is actually verified

Paper equation used by the hand derivation

For $z < 1$ and $h > 0$, the span of control is

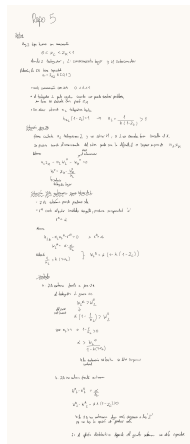
$$n(z) = \frac{1}{h(1-z)} \implies n(z)h(1-z) = 1.$$

Lean statement and proof fragment

```
theorem span_mul_helpCost_one_sub
  (E : Economy) (z : Real) (hz : z < 1) :
  span E z * (E.helpCost * (1 - z)) = 1 := by
  rw [span]
  exact one_div_mul_cancel
    (mul_ne_zero (ne_of_gt E.helpCost_pos)
      (sub_ne_zero.mpr (ne_of_gt hz)))
```

Domain encoded: $z < 1$ and the structure assumption $0 < h$; unlike the paper's human domain, it does not also require $0 \leq z$. The official fast check builds successfully. This verifies the identity, **not** the continuum equilibrium or Propositions 5–6.

Where I did not believe the AI



AI gloss

“Distributional effects are driven by autonomy, not capability.”

Audit verdict: incomplete

Prop. 5 requires $z_{AI} > \bar{z}_{AI}$ for autonomous bottom winners. Prop. 6 requires $z_{AI} > w(0)$ before non-autonomous AI is used.

- “Top always wins” additionally requires $h < h_0$ and $z_{AI} < 1$.
- $Y^* > Y^*$ assumes identical abundant compute and two layers.
- The two-type calculation is an extension, not the full market-clearing proof.

Takeaways

1. The model's mechanism is organizational matching and surplus division inside knowledge hierarchies.
2. Autonomous AI produces top winners under $h < h_0$, but bottom winners require $z_{AI} > \bar{z}_{AI}$.
3. Non-autonomous AI helps the bottom most only weakly when inactive and strictly when $z_{AI} > w(0)$.
4. Autonomy raises total output by letting compute pursue opportunities; non-autonomy can transfer more surplus to low-skilled users.

One-sentence answer

Capability decides whether AI is good enough to activate the distributive channel; autonomy decides which channel operates and who captures it.

github.com/johnbarraza/ai-05-ide